Using AI to Develop and Evaluate Module Student Learning Outcomes and Course Material

ILTA EdTech Conference 2024

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Background and Motivation

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- **AI in Education and Educational Technology**: Both Interested in AI in Education. Teacher and Learner Perspectives (John and Mark's Research & Mark's PhD).
- **Quality Measures in Education**: Interested in Quality Measures in Module Design. Student Learning Outcomes (SLOs) (John's Role as ADTL for FSE).
- **Learning Educational (Qualitative) Research using AI**: Investigating how to utilise AI to help learn qualitative research methods (John's interest in teaching Mark about qualitative research; Mark's Learning new methodologies utilising AI).
- **Designing and Implementing AI- based Software Solutions**: Investigating advanced prompting and utilising prompting frameworks (Marks's PhD research).

The Project and Learning Exercise!

- **Student Learning Outcomes (SLOs)**: As part of John's work as ADTL in the Faculty of Science & Engineering (FSE) he is interested in establishing the Quality of MU (FSE) Modules. One aspect of this work is to establish the quality of the SLOs for each module offered by the Faculty. What is a Quality module? What is a "good" SLO? Are they Learning Outcomes or Learning Objectives? Can you independently determine (evaluate) the quality of something using heuristics? Can AI help?

- **We were able to get the data**: We thought of all the many ways to do this - scrape (Python) - etc. but in the end we just asked Colleagues in the Curriculum (Registry) office.

- **Learning Opportunity for Mark**: We realised that this could be a classic Inter-Rater Reliability learning example. And it would be cool to have AI raters involved here!

Playing with AI and Course Development

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- **Generated Courses using AI**: We achieved this by using our CRTRF Framework (prompting framework) to generate courses together with a GPT-generated rubric for evaluating the courses.

- **Learning Opportunity - AI on SLO Evaluation**: We engaged in an extensive dialogue with AI (GPT3.5) on how to evaluate SLOs. Not just simple prompts - had a dialogic-like interaction with AI; reflection and correction. Eventually determined a way to evaluate SLOs in a systematic way. Still depends on a rater to evaluate each using rubric.

- **Generated SLOs for the Courses**: We later worked with AI to generate 100s of SLOs from module descriptors. We wanted a range of quality SLOs and AI did not want to help with this initially - wanted to give something it considered "the best".

PROJECT STEPS

- **DATA**: Decided to have two sources of data. Actual data collected from MU Registry and synthetic data generated by AI. AI generated data would include Module Descriptors, Schedules, SLOs, and Assessment Recommendations.

- **QUALITY EVALUATION**: The Quality Evaluation (presented here) focuses on SLO evaluation only. We focus on Mark and John's taught CS UG Modules mostly.

- **EdTech CONTEXT**: Using AI technology to learn about AI use in EdTech and to learn about Qualitative Methods in EdTech.

- **CRITICAL THINKING:** Mixture of Socratic (Mark) Interaction (a shared dialogue between teacher and students. The teacher leads by posing thought-provoking questions. Students actively engage by asking questions of their own with the discussion goes back and forth) and Dialogic (John) Interactions (listening to each other, explain thinking, and engage in sustained discussion of complex idea). And ... yes ... we treat AI like another human in these processes.

QUESTIONS?

- **QUESTION**: Can AI generate high quality Course Content and SLOs?
- **QUESTION**: Can AI help plan and learn Qualitative Research?
- **QUESTION**: Can AI help teach the Inter-Rater Reliability methodology?
- **QUESTIONS**: Can AI perform reliable SLO Ratings?

These are the core questions of interest for this project.

What are the **alternatives** to learning about Qualitative Research in EdTech **without using AI**? Read lots of papers, learn about statistics (not necessarily in context). Engage in dialogic interactions with a busy PhD supervisor. Use statistics software. Write programs, etc., etc. Can share prompts (dialogue), edit prompts and keep records.

Can AI generate Course Content?

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- **COURSE CONTENT**: Yes - AI does a great job here. You need to supply contexts. Mark previously developed his **CRTRF** framework for prompting. We used his framework here.

- **CONTEXT:** *The following is the data for an Introductory Computer Science course {data}.* **ROLE:** *As a professor of Computer Science…* **TASK:** *you will generate course syllabi materials and assessment strategies for the provided set of existing learning objectives.* **REQUIREMENTS:** *You must discuss the reasoning behind your choices and mention the topics that the educator should cover in their class.* **FORMAT:** *Please format your output as a Markdown document.*

- **REPRODUCIBILITY:** Across LLMs / Apps.

Context:

The following are the module name, year of study, student learning o
the number of credits for the module.

Module Name: {name}
Year of Study: {year}
Student Learning Outcomes: {learning_outcomes}
CA/Exam Ratio: {ca_exam_ratio}
Number of Credits: {year_credits}

100 Modules Generated
Different LLMs, etc.

Role Generation:

Take a deep breath and work on this problem step-by-step. You will act as a professor of Computer Science with several years of experience teaching {name}.

Task:

As a professor teaching {name}, you are tasked with creating a 12-week course syllabi.

Requirements:

This syllabus will include 12 detailed lecture topics, subtopics, and a detailed form of assessment for the week. You will also make sure the topic is relevant to both what areas are desired in the job market and what topics are foundational to current areas of research. Please also provide a reason for why the form of assessment you chose was selected using expert educational reasoning. You will also provide an improved course overview, five learning objectives for the course, a 12-week schedule for assignments, two full wrote sample lectures with subtopics and two sample assessment documents, where the assessment documents include the questions and tasks the students need to perform. Please be verbose in your answers.

Format:

Please provide this resulting text in a JSON format without any accompanying text.
Please follow the below format for the JSON file:

"Course": "Course Name",
"Module Description": "Module Description" of type string,
"Year of Study": "Year of Study" of type string,
"Learning Objective": "Learning Objective" of type string,
"CA/ Exam Ratio": "CA/ Exam Ratio" of type string,
"Course Credits": "Course Credits" of type string,

"Course Overview": "Course Overview" of type string,
"Learning Objectives": "Learning Objectives" of type array of strings,
"Schedule": "Schedule" type array of strings,
"Lectures": "Topics and Subtopics" of type object,
"Assessments": "Assessments" of type object,
"Sample Lectures": "Sample Lectures" of type object,
"Sample Assessments": "Sample Assessments" of type object

Using JSON for adding
to MongoDB for
Evaluation

Once again, please ensure the output is a valid JSON object with no accompanying text.

Can AI generate and Evaluate SLOs?

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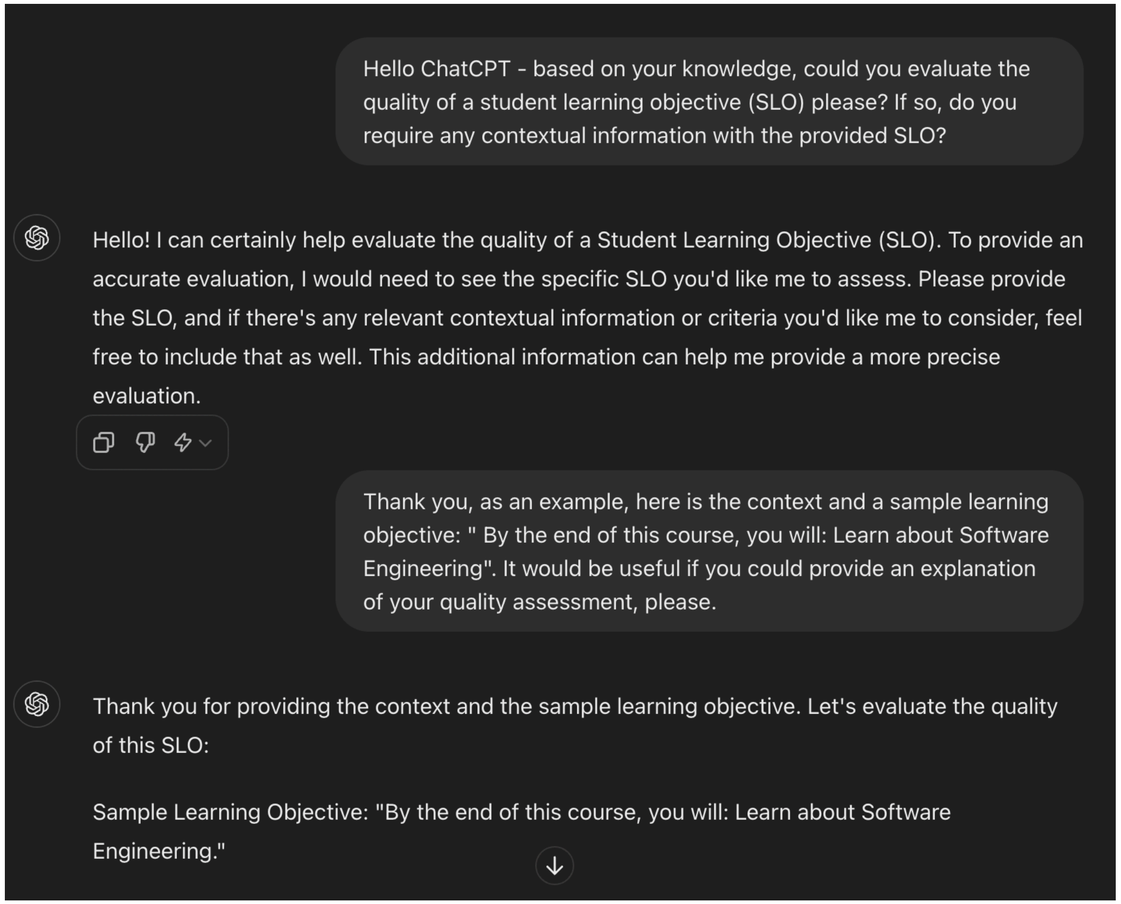
- **GENERATE SLOs**: **Yes** - AI does a great job here teaching about SLOs and approaches to evaluation. All using specific examples. You need to supply contexts and to critically engage with responses.

- **EVALUATION METHODS**: **Yes** - AI does a great job explaining the SMART system and also presenting a model for evaluating SLO Quality.

- **INTRA-REFERENCING:** AI does great job referring to earlier results. Prompts do not have to exist in isolation. Be Critical. Have done this with **ChatGPT**, **Bing Chat/ Copilot**, **Google Gemini**, **Claude**.



Start of First Discussion with AI on SLOs (100+ prompts)

SMART SLOs (AI Recommendation)

- **AI RECOMMENDATION**: Using the SMART approach to SLO generation and evaluation. Provides ranking for individual SLOs under the SMART headings. We know this already, of course, but it is great to see this from AI.

- **EVALUATION CONTEXT**: This gives AI and Human Raters a common scale to use when evaluating SLOs.

- **PROMPT**: Can use this information with LLMs (via Prompts) to perform rating and produce appropriate results tables.

- **PROMPT**: Can use this information to generate SLOs for specified Module Descriptors (CRTRF).

GPT-3.5 Generated LO Evaluation Scale (SMART)
Specificity / Clarity: 0: Vague or unclear SLO with no specific focus. 1: SLO is somewhat clear but could be more specific. 2: Clear and specific SLO that clearly states what students are expected to learn.
Measurability: 0: SLO lacks a measurable action verb. 1: SLO includes a measurable verb but is still somewhat vague. 2: SLO includes a clear and measurable verb, making it easy to assess.
Achievability: 0: SLO is unrealistic or unattainable within the course context. 1: SLO is somewhat attainable but may be challenging. 2: SLO is realistic and achievable within the course.
Relevance: 0: SLO is not relevant to the course's objectives or content. 1: SLO is somewhat relevant but could be more closely aligned. 2: SLO is highly relevant and directly related to the course content.
Timeframe: 0: No specified timeframe for achieving the SLO. 1: Timeframe is vaguely defined. 2: Clear timeframe specified for achieving the SLO.

Generating Synthetic Data for Learning

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Certainly, I can provide you with an example of how you can compute statistics and present them in a table for a research paper using the quality assessment approach for a set of randomly generated SLOs for a module on Software Engineering.

Let's say you have evaluated 20 SLOs using the quality scale and scoring system I provided earlier, and you want to present the statistics in a table. Here's a sample table:

plaintext Copy code

```
| SLO Number | Clarity | Measurability | Achievability | Relevance | Timeframe | T
|------------|---------|---------------|---------------|-----------|-----------|---
| SLO 1      |    2    |       2       |       2       |     2     |     2     |
| SLO 2      |    1    |       1       |       2       |     2     |     2     |
| SLO 3      |    2    |       2       |       1       |     2     |     2     |
| SLO 4      |    1    |       2       |       2       |     1     |     2     |
| SLO 5      |    2    |       1       |       2       |     2     |     1     |
| SLO 6      |    1    |       1       |       1       |     2     |     2     |
| SLO 7      |    2    |       2       |       2       |     2     |     1     |
| SLO 8      |    1    |       1       |       1       |     1     |     2     |
| SLO 9      |    2    |       2       |       2       |     2     |     2     |
| SLO 10     |    2    |       2       |       1       |     2     |     1     |
| SLO 11     |    1    |       1       |       1       |     2     |     1     |
| SLO 12     |    2    |       2       |       2       |     2     |     1     |
| SLO 13     |    1    |       2       |       1       |     1     |     2     |
| SLO 14     |    2    |       2       |       2       |     2     |     2     |
| SLO 15     |    1    |       1       |       2       |     1     |     1     |
| SLO 16     |    2    |       2       |       1       |     2     |     1     |
```

AI Play - Investigating Synthetic SLO Statistics (ChatGPT)

I can generate a list of 50 randomly-generated Learning Objectives (LOs) for a module on "Data Structures & Algorithms" and tag each LO with a randomly selected quality measure (poor, moderate, or high). Here's a table with the LOs:

LO Number	Learning Objective	Quality
LO 1	Implement common data structures like arrays, linked lists, and stacks.	High
LO 2	Understand the concept of time complexity and analyze the time complexity of algorithms.	High
LO 3	Identify appropriate data structures for specific problem-solving scenarios.	High
LO 4	Implement tree structures, including binary trees and AVL trees.	High
LO 5	Explain and apply fundamental sorting algorithms like insertion sort and selection sort.	Moderate
LO 6	Solve problems involving recursion and demonstrate a solid understanding of recursive algorithms.	Moderate
LO 7	Analyze the space complexity of algorithms and optimize memory usage.	Moderate
LO 8	Design and implement hash tables and understand their applications in data storage.	Moderate
LO 9	Demonstrate proficiency in using dynamic programming to solve complex algorithmic problems.	High
LO 10	Evaluate and compare different s ching algorithms for efficiency and effectiveness.	Moderate

AI Play - Generating SLO of varying Quality (ChatGPT)

Generating Synthetic Data for Learning

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Absolutely! Here's a table evaluating the SLOs you provided using the same rubric and referencing the course description:

Measurability (0-20)	Achievability (0-20)	Relevance (20-20)	Timeframe (20-20)	Total Score
0 (Unclear how to assess)	8 (Possible, but very broad)	20 (Relevant)	0 (No timeframe)	32
10 (Assessment through exams/quizzes)	18 (Likely achievable with effort)	20 (Relevant)	14 (Should specify by semester/course end)	78
4 (Unclear how to assess discussion)	10 (Possible, but depends on discussion depth)	10 (Somewhat relevant)	0 (No timeframe)	32
15 (Assessment through presentations/written assignments)	18 (Likely achievable with good understanding)	20 (Relevant)	15 (Should specify within a quarter)	86

Enter a prompt here

AI Play - Rating Synthetic SLO Evaluations (Google Gemini)

Okay - now I would like you to evaluate the follo...

File Edit View Insert Format Data Tools Extensions Help

A1 | LO Number

	A	B	C	D	E	F	G	H	I
1	LO Number	Learning Objecti	Quality	Clarity (0-20)	Measurability (0-	Achievability (0-2	Relevance (20-2	Timeframe (20-2	Total Score
2	SLO 1 (Poor)	Learn data struc	Poor (32)	4 (Vague and lac	0 (Unclear how t	8 (Possible, but	20 (Relevant)	0 (No timeframe	32
3	SLO 2 (Moderate	Understanding a	Moderate (64)	12 (Clear on unc	10 (Assessment	18 (Likely achiev	20 (Relevant)	14 (Should spec	78
4	SLO 3 (Moderate	Discuss Progran	Poor (40)	8 (Unclear what	4 (Unclear how t	10 (Possible, bu	10 (Somewhat re	0 (No timeframe	32
5	SLO 4 (Good)	Explain basic da	Good (76)	18 (Clear on exp	15 (Assessment	18 (Likely achiev	20 (Relevant)	15 (Should spec	86
6	SLO 5 (Good)	Implement simpl	Good (88)	20 (Clear on imp	20 (Measurable	18 (Likely achiev	20 (Highly releva	18 (Should spec	96
7	SLO 6 (Moderate	Describe the cor	Moderate (72)	15 (Clear on des	15 (Assessment	18 (Likely achiev	20 (Relevant)	14 (Should spec	84
8	SLO 7 (Good)	Design and impl	Good (92)	20 (Clear on des	20 (Measurable	20 (Likely achiev	20 (Highly releva	22 (Should spec	112
9	SLO 8 (Good)	Analyze and con	Good (92)	20 (Clear on ana	20 (Measurable	18 (Likely achiev	20 (Highly releva	20 (Should spec	108
10	SLO 9 (Good)	Develop and tes	Good (90)	20 (Clear on dev	20 (Measurable	18 (Likely achiev	20 (Highly releva	22 (Should spec	100
11									
12									

AI Play - Exporting Synthetic Data from Google Gemini

- **EVALUATION**: **Yes** - AI (all of them) does a great job evaluating SLOs against criteria. AI may actually be excellent Raters. Can always ask for "reasoning" too! Good Practice!

Can AI generate high quality SLOs?

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- **QUALITY SLOs**: No - AI does not do a great job here. You have to work really hard to make AI generate POOR or MODERATE quality SLOs. It does a great job on HIGH quality.

- **BEST:** AI wants to do a good job (it would appear). Even the POOR quality generation using the Framework usually gives great SLOs.

- **URELIABLE**: We have generated SLOs of varying quality and then used the same (and different AIs) to rate quality. AI rates its own generated SLOs inconsistently, e.g. rates MODERATE as GOOD, etc.

Certainly, here's a list of 25 randomly-generated Learning Objectives (LOs) for a module on "Data Structures & Algorithms," with each LO tagged as "POOR" quality. The LOs are also assessed for clarity, measurability, achievability, relevance, timeframe, and total score to illustrate the reasons for their poor quality:

LO Number	Learning Objective	Quality	Clarity	Measurability	Achievability	Relevance	Timef
LO 1	Learn stuff about data structures.	POOR	Poor	Poor	Poor	Poor	Poor
LO 2	Understand algorithms for things.	POOR	Poor	Poor	Poor	Poor	Poor
LO 3	Explore a bit of data structures and things.	POOR	Poor	Poor	Poor	Poor	Poor
LO 4	Know some data structures but not really.	POOR	Poor	Poor	Poor	Poor	Poor
LO 5	Be aware of algorithms and stuff.	POOR	Poor	Poor	Poor	Poor	Poor

AI Play - Generating POOR Synthetic SLOs (ChatGPT)

Qualitative Research: Inter-Rater Reliability

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- **Inter-Rater Reliability (IRR)**: Inter-rater reliability refers to the degree of agreement among multiple raters or evaluators when assessing the same thing (quality of SLOs). It is used to determine the consistency of measurements when different people evaluate or measure the same subject or object. A high inter-rater reliability indicates that the assessment is consistent and reproducible across different raters.

- **Getting the SLO Data**: We thought of all the many ways to do this - scrape MU Court Finder (Python) - etc. but in the end we just asked Colleagues in the Curriculum (Registry) office! We secured all SLOs and Course Descriptors.

- **Opportunity for Learning**: We realised that this could be a classic Inter-Rater Reliability learning example. And it would be cool to have AI raters involved here! different AI raters, Cohen's Kappa, Coding, SLO Heuristics/Rubric, Visualisation, etc.

Can AI help you learn Qualitative Research?

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- **LEARNING ABOUT IRR**: AI does a fantastic job teaching about IRR in a specific context. We were able to work with SLOs and generate AI Raters for Synthetic and Real SLO data.

- **RATERS**: We implemented Prompt-based Raters using the CRTRF framework interactively in various LLMs. We implemented raters using LLM APIs (Python) for large data sets contained in databases.

- **IRR STATISTICS**: We were able to interactively work with AI to generate contingency tables, learn about formulae, perform calculations, generate code, query and discuss findings (**Cohen's Kappa** for pairs from three AI raters).

- **EXTRA STUFF**: Also worked with **Fleiss's Kappa** which considers the agreement beyond chance for multiple raters classifying items into nominal categories. And **Kendall's Coefficient of Concordance** (W) for Inter-Rater Reliability.

Learning Inter-Rater Reliability (K and W)

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We'll use the same ratings from Rater 1 (R1), Rater 2 (R2), and Rater 3 (R3) for the nine SLOs.

LO Number	R1 Score	R2 Score	R3 Score
1	M	P	G
2	M	M	G
3	P	P	M
4	G	G	G
5	G	G	G
6	G	M	G
7	M	M	F **(Invalid score)**
8	M	G	M
9	G	G	M

AI Play - Sample Data for Kappa play - Google Gemini

We'll compare the ratings for each pair of raters (R1 vs R2, R1 vs R3, R2 vs R3) for each item (excluding LO7).

LO Number	R1 vs R2 (S)	R1 vs R3 (S)	R2 vs R3 (S)
1	1 (R1 > R2)	-1 (R1 < R3)	-2
2	0 (Tie)	0 (Tie)	0 (Tie)
3	0 (Tie)	1 (R1 > R3)	1 (Difference of S values)**
4	0 (Tie)	0 (Tie)	0 (Tie)
5	0 (Tie)	0 (Tie)	0 (Tie)
6	1 (R1 > R2)	0 (Tie)	1 (Difference of S values)**
8	-1 (R1 < R2)	1 (R1 > R3)	0 (Difference of S values)**
9	0 (Tie)	1 (R1 > R3)	1 (Difference of S values)**

Export to Sheets

AI Play - Learning Kendall's W - IRR and AI SLO Raters (Google Gemini)

Can AI perform reliable SLO Ratings?

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AI SLO Rater Reasoning (ChatGPT)

Reasoning:

- **Clarity and Measurability:**
  - Most SLOs (4, 5, 6, 7, 9) use strong action verbs and clearly state what students are expected to learn (20 points).
  - SLOs 1 and 2 are too vague (15 points).
  - SLO 3 lacks a measurable verb and needs more specifics (5 points).
- **Achievability:**
  - Considering the introductory nature of the module ("Introduction to algorithms and data structures"), most SLOs are achievable within the course timeframe (15-20 points).
  - Complexity of analyzing sorting algorithms (SLO 8) and designing linked lists (SLO 7) might require adjustments (15 points).
- **Relevance:**
  - Most SLOs directly align with the module content (20 points).
- **Timeframe:**
  - Neither the module descriptor nor SLOs specify a timeframe (0 points).

Overall:

The module descriptor provides a good foundation for the course. The SLOs are generally well-written and relevant, but some could benefit from increased clarity and specifying a timeframe for achievement.

Recommendations:

- Revise SLOs 1 and 2 to be more specific (e.g., "Apply basic data structures like stacks and queues to solve problems").
- Consider adding a timeframe to the SLOs or the module description (e.g., "By the end of the module, students wi be able to...").
- Refine SLO 3 to include a measurable verb and specify the programming concepts to be discussed.

LO Number	Learning Objective	Quality	Clarity (0-20)	Measurability (0-20)	Achievability (0-20)	Relevance (0-20)	Timeframe (0-20)	Total Score
1	Learn data structures	1	15	10	15	15	0	65
2	Understanding algorithms	1	15	10	15	15	0	65
3	Discuss Programming	0	5	5	10	5	0	30
4	Explain basic data structures like stacks and queues	2	20	20	20	20	0	80
5	Implement simple sorting algorithms like bubble sort and quick sort.	2	20	20	20	20	0	80
6	Describe the concept of object-oriented programming and its principles.	2	20	20	20	20	0	80
7	Design and implement a linked list data structure to manage dynamic data.	2	20	20	15	20	0	75
8	Analyze and compare the efficiency of different sorting algorithms including selection, insertion, and quick sort.	2	20	20	15	20	0	75
9	Develop and test a binary search algorithm for ordered datasets.	2	20	20	20	20	0	80

AI SLO Rater (ChatGPT)

- **RELIABLE?**: AI Can Rate Well but we are unsure as to how reliable the ratings are (unfortunately) at this stage. They work well with Human Generated POOR, MODERATE and GOOD SLOs but not with synthetically generated SLOs.

Can AI help learn/plan Qualitative Research?

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- **LEARN QUALITATIVE RESEARCH**: Yes! AI, and a learning technology, provides an immersive and localised learning experience for learning about qualitative research.

- **CONTEXT**: For the IRR context at least! Will need to investigate other forms. What is of interest here is that it also provides opportunities for mixed approaches. We can use AI to help learn about Qualitative Analyses of Quantitative Data <u>and</u> Quantitative Analyses of Qualitative Data.

- **ETHICS**: When it is difficult to obtain real data (prior to ethical approval, etc.) or raters but you want to progress your research - AI can definitely help! You can perform extensive pilot research using AI. Learn to Work with Real data!

- **PLANNING**: When planning research AI can definitely help with the steps required!

Please fell free to email comments, questions, etc.:

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Thank you and Questions?